

flashjet

GPU jet clustering — benchmarking against FastJet

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What flashjet is

A GPU implementation of the standard sequential-recombination jet algorithms

— anti- k_t , k_t , Cambridge–Aachen — written in Triton/PyTorch.

- Clusters a **batch of jets or events at once**, directly in GPU memory.
- The data **never moves back to the CPU** during clustering.
- Returns the particle → jet assignment **and the complete merge history** (the full binary tree of which pair merged, and at what distance).

Why the merge history matters: substructure observables — groomed mass, z_g , R_g , Lund coordinates, exclusive subjets — are then just **cheap reads of that tree**, with no second clustering pass.

The goal of this study: is flashjet a viable **substitute for FastJet**?

That needs two things — it must give the **same jets**, and it must be **faster**.

What was established before this study

The physics was validated; the speed was not.

what	how it was checked	outcome
the algorithms are correct	independent NumPy tree-walks; 129 unit tests	pass
clustering matches FastJet	jet-by-jet kinematics tests	exact
works on real detector data	reclustered CMS jets vs the values CMS stored	p_T ratio 1.000000
grooming is correct	our soft-drop mass vs CMS's	median Δ -0.004 GeV
substructure is correct	R_g, z_g vs CMS	$\Delta \leq 2 \times 10^{-4}$
the physics is right	soft-drop z_g , Lund plane, jet areas vs analytic predictions	reproduced

Small disagreements were traced and explained rather than ignored: they come from how the input file **stores** numbers (dropped soft particles, float rounding), not from the algorithm.

⇒ **Correctness was settled. What was missing was a proper answer to "how much faster is it, on real physics data?"** — that is this study.

The benchmark

what · on what · how

What is being compared

Two clusterers, given **exactly the same jets**, asked to do **exactly the same job** (anti- k_t , same radius R):

	runs on	what it is
flashjet	GPU	this library
FastJet — "vectorised"	CPU	the standard tool, called through its efficient batched interface
FastJet — "per-jet"	CPU	the standard tool, called one jet at a time

All speedups quoted here are against the vectorised (faster) FastJet.

What data

CMS Open Data — publicly released simulation, so every number here can be shown freely, with no approval needed.

process	jets used	average particles per jet
DY+jets	30 000	10.8
W+jets	30 000	15.1
$t\bar{t}$ (all-hadronic)	30 000	22.2
QCD (high- p_T dijets)	28 250	34.0

Format: **MINIAOD**.

How the measurement is done

- 1. Take the jets.** Extract the particles belonging to each jet from the Open Data files, once, and save them.
- 2. Cluster with flashjet on the GPU.** Time it, then **save the exact same input events to a file.**
- 3. Cluster those same saved events with FastJet on the CPU.** Time it.
- 4. Check they agree.** Compare the **number of jets found**, event by event. If the two disagree, the timing is meaningless.

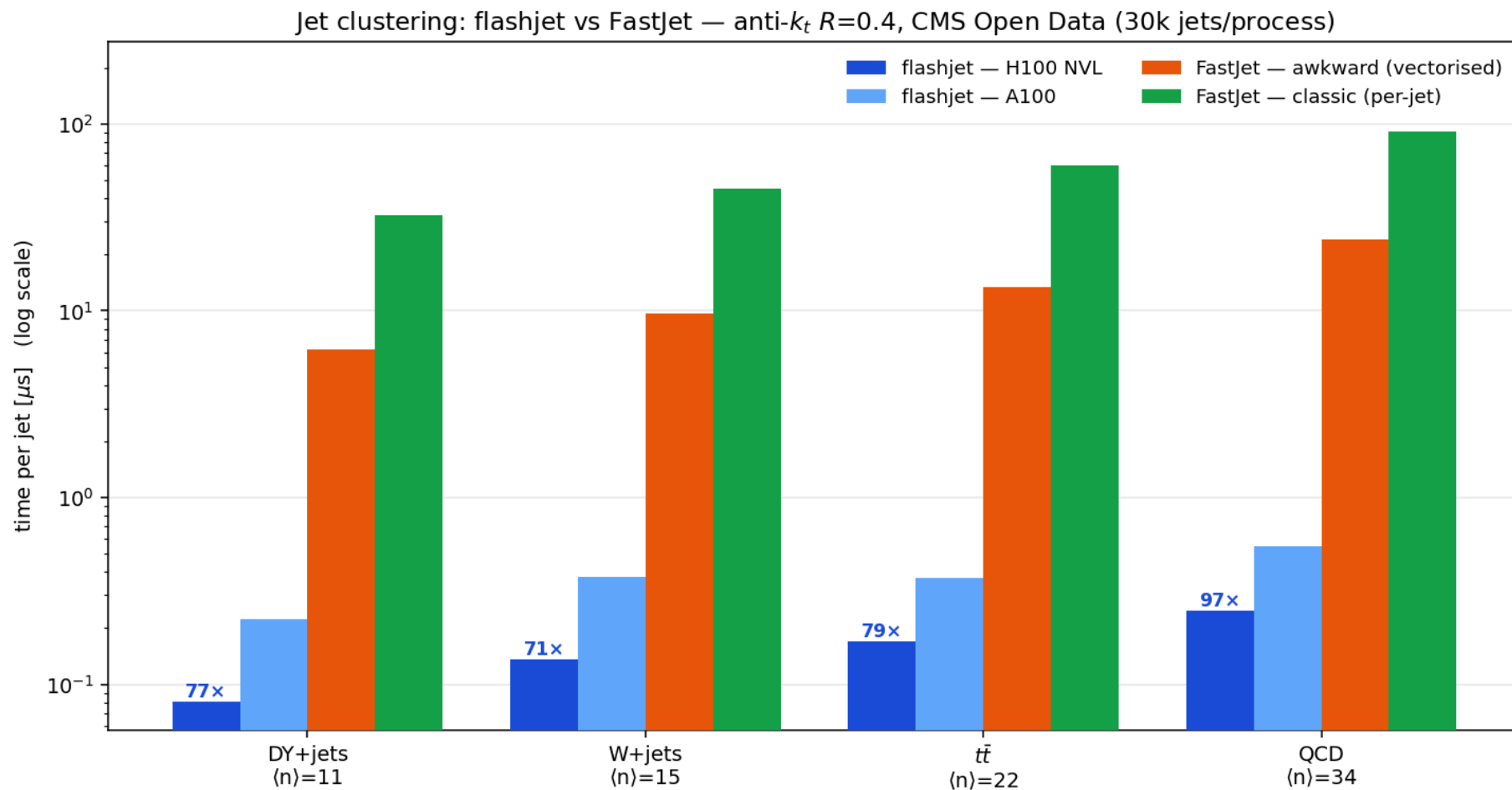
Repeated for **4 processes × 2 jet radii ($R = 0.4$ and 0.8) × 2 GPUs.**

The key discipline: both clusterers read the **same saved file**. Nothing is regenerated in between, so this is a like-for-like comparison and not two separate measurements that happen to use the same dataset name.

Hardware: NVIDIA **A100** and **H100 NVL**, one full GPU each (no shared or partitioned cards). CPU baseline runs on the same machine.

Results

Time to cluster one jet



Anti- k_t , $R=0.4$. **Log scale** — each gridline is 10 $\times$. Blue = flashjet on GPU, orange/green = FastJet on CPU. Labels give the speedup over vectorised FastJet.

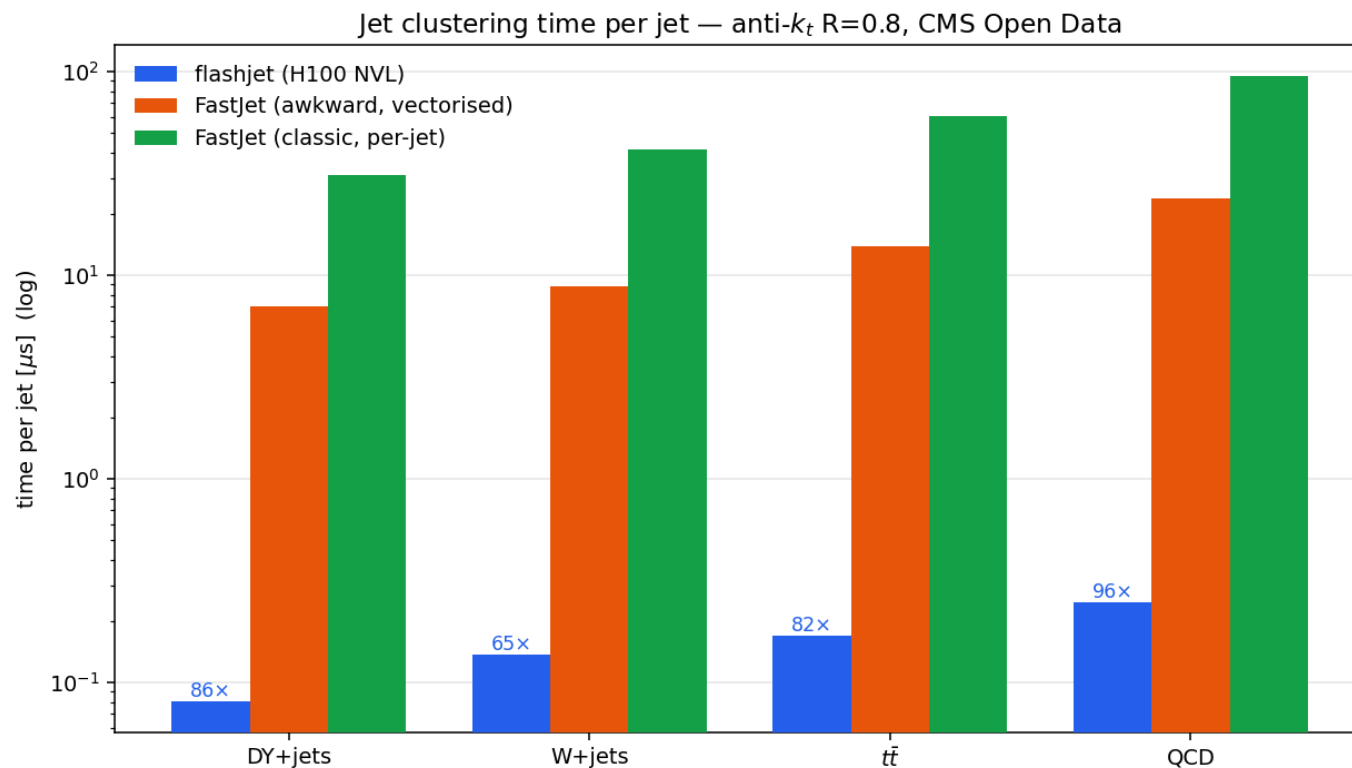
The numbers

Microseconds to cluster **one jet** (anti- k_t , $R=0.4$):

process	particles/jet	flashjet H100	flashjet A100	FastJet (vectorised)	FastJet (per-jet)	speedup
DY+jets	10.8	0.081	0.223	6.20	32.43	77×
W+jets	15.1	0.137	0.373	9.67	44.84	71×
$t\bar{t}$	22.2	0.170	0.373	13.46	59.95	79×
QCD	34.0	0.249	0.549	24.24	91.64	97×

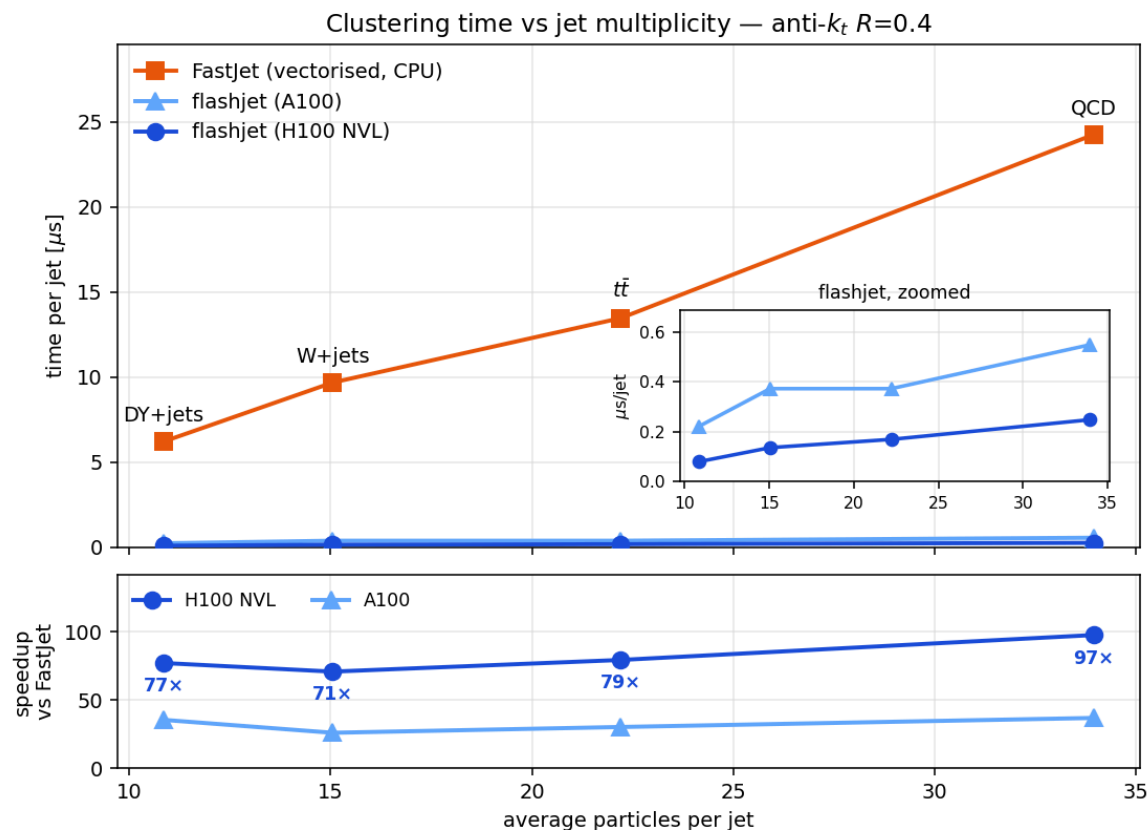
- **Same jets:** number of jets found agrees with FastJet in **~100 %** of cases (7 of 8 runs exactly 100 %, one at 99.997 %).
- **Not a fluke of the jet radius:** $R=0.8$ gives the same picture (H100 65–96×).
- **Newer GPU helps:** A100 → H100 is a further ~2.2× on identical input.

The same holds at a larger jet radius ($R = 0.8$)



Anti- k_t with $R=0.8$ instead of 0.4: **65–96x** over vectorised Fastjet on H100 (A100: 25–41x), again with ~100 % agreement on the number of jets found. **The conclusion does not depend on the choice of jet radius.**

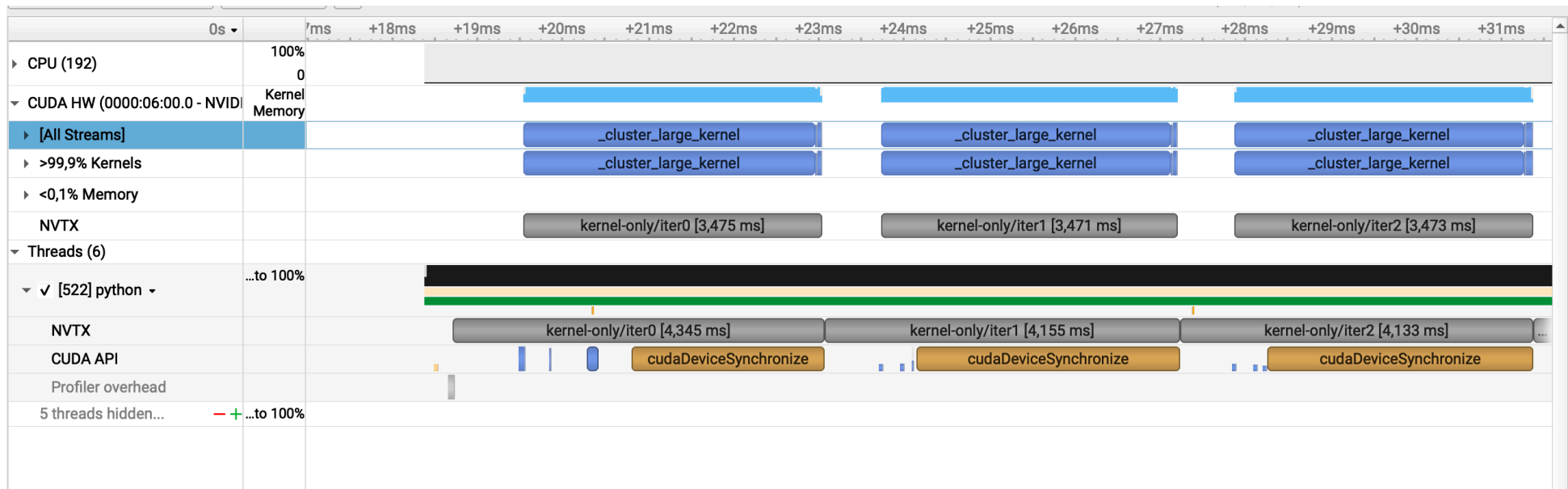
The trend that matters



Linear scale. **Top:** time per jet — FastJet's cost climbs steeply with jet multiplicity while both flashjet curves stay nearly flat (inset zooms in on them).

Bottom: the resulting speedup. The four processes sit on **one common trend**, so this is a property of the algorithm rather than of any particular sample.

What the GPU is actually doing



Nsight Systems timeline, H100, three consecutive iterations. Each blue block is **one clustering call**; the rows are labelled by the tool itself as ">99.9 % Kernels" and "<0.1 % Memory".

The kernels run back to back with nothing in between. The memory row is **empty** — no data is shuffled between CPU and GPU while clustering runs. That is the property that makes the speedup real rather than an artefact of where the timer was started.

Profiling — where the time goes, and what limits it

Nsight Systems — share of GPU time (A100)

kernel	% GPU	avg / call
<code>_cluster_large_kernel</code>	97.6 %	5.47 ms
<code>_decode_kernel</code>	1.5 %	82 μ s
torch glue	< 1 %	—

CPU $\leftrightarrow$ GPU transfer negligible once running.

⇒ **No hidden overhead** — essentially all the time is the real clustering work.

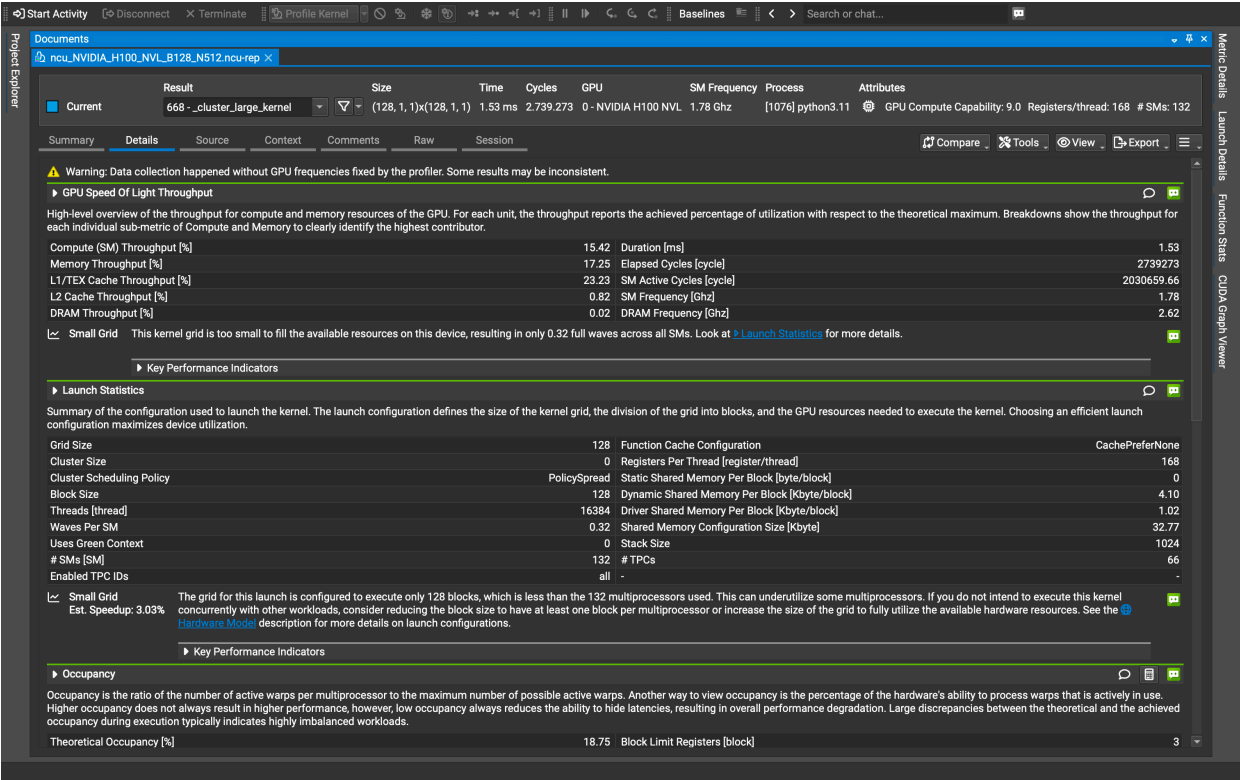
Nsight Compute — hardware counters (H100)

metric	value
DRAM throughput	0.02 %
Compute (SM) throughput	15.4 %
Registers / thread	168
Theoretical occupancy	18.75 %
Achieved occupancy	6.25 %
Waves per SM	0.32

⇒ **Not memory- or compute-bound** — the GPU is simply **not filled** (0.32 waves/SM).

flashjet is already ~80× faster while leaving most of a modern GPU idle — the work-distribution settings were tuned on an older, smaller card. **Headroom, not a limit.**

The profiler names the bottleneck



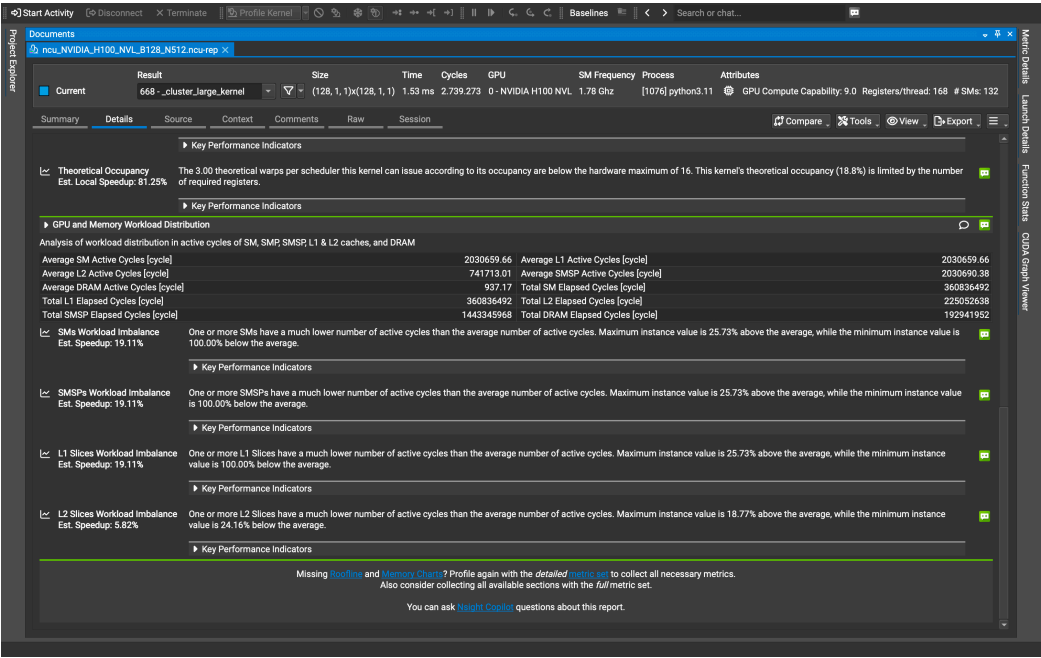
"Small Grid" — "This kernel grid is *too small to fill the available resources on this device, resulting in only 0.32 full waves across all SMs.*"

Grid size = 128 blocks but # SMs = 132 — the kernel launches **fewer blocks than the GPU has processors**, so some are guaranteed to sit idle.

Nsight Compute, Speed-of-Light + Launch Statistics — the tool's own words and numbers.

flashjet — GPU jet clustering: benchmarking

...and prices the fix



The tool's own speedup estimates:

finding	est. speedup
Theoretical occupancy (registers)	81 %
Achieved occupancy	67 %
SM workload imbalance	19 %
SMSP / L1 imbalance	19 %

*"...theoretical occupancy (18.8 %) is **limited by the number of required registers.**"*

*"One or more SMs have a much lower number of active cycles... **minimum instance value is 100 % below the average.**" → some SMs do no work at all.*

⇒ ~1.7–1.8× more is available, by NVIDIA's own estimate — from launch configuration and register budget, not from the algorithm.

Conclusions

Conclusions

1. **flashjet gives the same jets as FastJet.**
~100 % agreement on jet counts across every process and radius tested, on top of the earlier jet-by-jet agreement with real detector data.
2. **flashjet is 65–97× faster** than the standard vectorised FastJet on a modern GPU — **0.08–0.25 μ s per jet.**
3. **The advantage grows with jet complexity.** Busier jets → bigger speedup. This is the regime that matters most for boosted-object physics.
4. **The result is robust**, not tuned: four unrelated physics processes, two jet radii, two GPU generations, all consistent.
5. **There is still significant headroom.** The GPU is largely idle, and the profiler attributes this to register pressure — it estimates a further ~1.7–1.8× is available from that alone.

Bottom line: as a drop-in replacement for CPU jet clustering, flashjet is both correct and roughly two orders of magnitude faster.

What is done, what is next

Checklist

Correctness — complete

- ✓ 129 unit tests pass, including GPU-only tests
- ✓ matches FastJet exactly in kinematics tests
- ✓ matches real detector-stored values jet-by-jet (p_T , groomed mass, R_g , z_g)
- ✓ reproduces analytic physics predictions
- ✓ same number of jets as FastJet in every timing run

Performance — complete

- ✓ 4 physics processes $\times$ 2 radii $\times$ 2 GPU generations, on public data
- ✓ profiled: no hidden overhead; bottleneck identified

Infrastructure — complete

- ✓ reproducible pipeline from public data to final plots
- ✓ results are freely presentable (no approval needed)

To do — in priority order

	item	why
1	Compare against C++ FastJet	every number here uses FastJet's Python interface. The production tool is C++. Until we measure it, the honest caveat is <i>"some of this could be Python overhead"</i>
2	Improve GPU utilisation	the profile says most of the GPU is idle; retuning how work is spread should give more speed for free
3	Other algorithms & conditions	only anti- k_t is timed (k_t , C/A untested for speed); no scan over jet radius; no test vs pile-up
4	Use it inside the experiment's software	the real production test. Needs the GPU code rewritten in C++ — a separate project

Backup

Reproducing this

```
# 1. public data -> per-jet particle lists
python3 dump_opendata_constituents.py {ttbar|qcd|wjets|dyjets} 30000 ak4

# 2. timing + profiling on a GPU (batch job)
condor_submit bench_a100.sub          # or bench_h100.sub

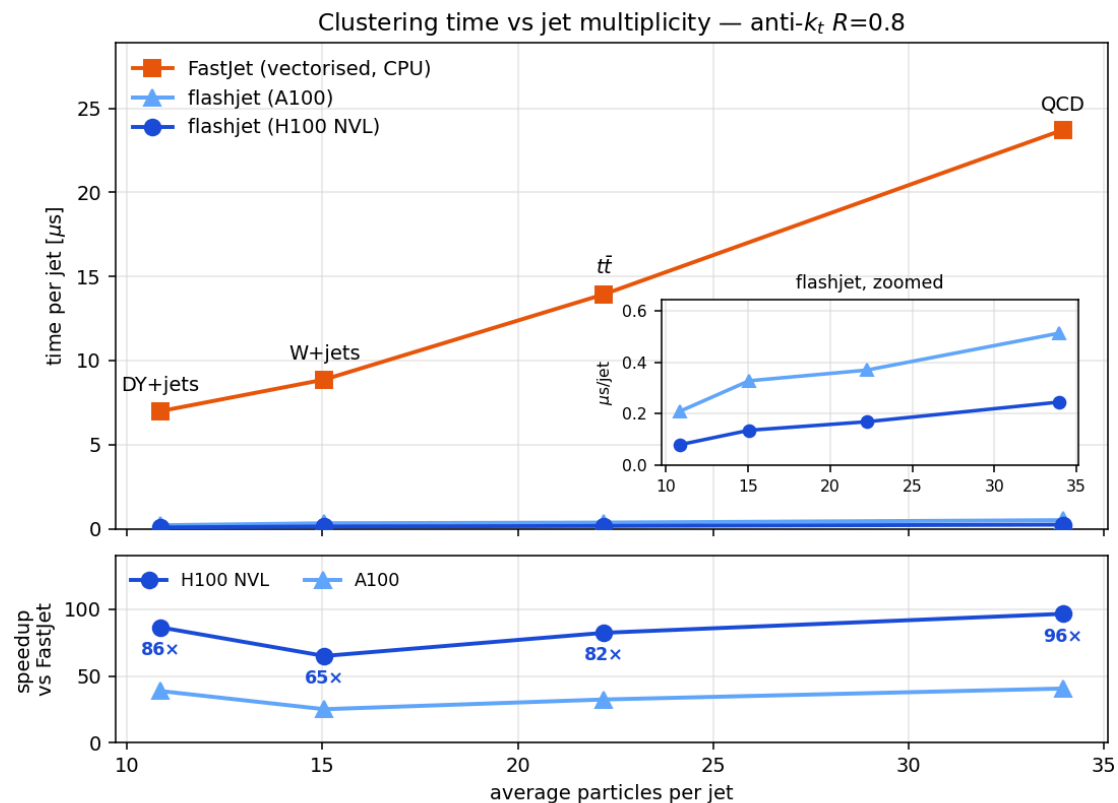
# 3. plots
python3 make_bench_plots.py 0.4 H100_NVL
```

Datasets (CMS Open Data, RunIISummer20UL16 MINIAOD):

process	dataset
$t\bar{t}$	TTToHadronic_TuneCP5_13TeV-powheg-pythia8
QCD	QCD_Pt_470to600_TuneCP5_13TeV_pythia8
W+jets	WJetsToLNu_1J_TuneCP5_13TeV-amcatnloFXFX-pythia8
DY+jets	DYJetsToLL_M-50_TuneCP5_13TeV-madgraphMLM-pythia8

Read directly over the network from public CERN storage — no grid certificate required.

$R = 0.8$ — multiplicity trend



Same construction as the $R=0.4$ trend slide, for the larger jet radius.

Reading the profiling numbers

counter	meaning	flashjet on H100
DRAM throughput	how hard memory is being pushed	0.02 % → memory idle
Compute (SM) throughput	how hard the arithmetic units work	15.4 % → not compute-bound
Registers per thread	resources each thread reserves	168 → high
Theoretical occupancy	how many threads <i>could</i> run, given the above	18.75 % → capped by registers
Achieved occupancy	how many actually did	6.25 %
Waves per SM	how many times the GPU was filled	0.32 → not filled once

The limitation is parallelism, not memory or arithmetic. Two concrete levers:

- 1. **Launch geometry** — the kernel launches **128 blocks on a 132-SM device**, so it cannot fill the GPU by construction, regardless of anything else.
- 2. **Register budget** — 168 registers/thread caps how many warps can be resident, which is what the profiler flags as limiting theoretical occupancy.

Both are configuration inherited from smaller GPUs, not properties of the algorithm.